		Statement 3: False. A chemical reaction produces a new radioactive compound but the nuclei are unchanged. Thus, half-life is unchanged.
30	C	Initial $A_0 = \lambda N_0 = (\ln 2 / 4 \text{ years})N_0 = 0.1733\ N_0$ After 2 years, $A_1 = \lambda N_1 = 0.1733(N_0 e^{-\lambda(2)}) = 0.1225\ N_0$ For the same treatment, (Duration x Activity) now and 2 years later should be equal. Hence, $(10 \text{ mins})(A_0) = (t)(A_1)$ $(10)(0.1733\ N_0) = (0.1225\ N_0)(t)$ $t = 14$ min

End of solutions